STA2002: Probability and Statistics II
Midterm Exam

March 24, 2021

The questionnaire comes with three questions. You need to answer them all. The number of points each question carries is stated next to it. The total is 100. You have 80 minutes in order to do so. You are entitled to bring along to the exam class four A4 sheets of papers which you have prepared in advance. Finally, you are asked to present a valid Student I.D. Card for inspection during the examination session.

1. For $n \geq 1$, let X_i, $1 \leq i \leq n$, and Y_i, $1 \leq i \leq n+1$, be two series of numbers. Moreover, $Y_i = X_i$, $1 \leq i \leq n$. A few claims are posed below. For each of them state whether it is true or false. In the former case, supply a proof. In the latter case, construct a counter example.

   (a) (6) $(X_i - \overline{X})^2 \leq (X_i - \overline{Y})^2$, $1 \leq i \leq n$

   (b) (6) $\Sigma_{i=1}^n (X_i - \overline{X})^2 \leq \Sigma_{i=1}^n (X_i - \overline{Y})^2$

   (c) (6) $\Sigma_{i=1}^n (X_i - \overline{X})^2 \leq \Sigma_{i=1}^{n+1} (Y_i - \overline{Y})^2$

   (d) (6) $\mathrm{Var}(X) \leq \mathrm{Var}(Y)$

   Solution:

   (a) False. A counter example is as follows: $n = 2$, $X_1 = 1$, $X_2 = 3$, $Y_3 = -1$. Then, $\overline{X} = 2$, $\overline{Y} = 1$ and $(X_1 - \overline{X})^2 = (1-2)^2 = 1$. On the other hand, $(X_1 - \overline{Y})^2 = (1-1)^2 = 0$, which is smaller.

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(b) True. It was proved in class

$$\overline{X} = \arg\min_x \sum_{i=1}^{n} (X_i - \overline{X})^2.$$

(c) True. Proof:

$$\sum_{i=1}^{n} (X_i - \overline{X})^2 \leq \sum_{i=1}^{n} (X_i - \overline{X})^2 = \sum_{i=1}^{n} (Y_i - \overline{X})^2$$

$$\leq \sum_{i=1}^{n} (Y_i - \overline{X})^2 + (Y_{n+1} - \overline{)}^2 = \sum_{i=1}^{n+1} (Y_i - \overline{X})^2.$$

(d) False. A counter example is as follows: $n = 2$, $X_1 = 1$, $X_2 = 3$ and hence $\overline{X} = 2$ and $\mathrm{Var}(X) = 1$. Add now $Y_3 = 2$. The mean $\overline{Y}$ stays 2 but

$$\mathrm{Var}Y = \frac{1}{3}[(1-2)^2 + (2-2)^2 + (3-2)^2] = \frac{2}{3},$$

which is smaller than the original variance of one.

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2. (20) Recall that if the regression line of Y on X, is $y = ax + b$, then the fitted value for individual i, is $\hat{Y}_i = aX_i + b$, $1 \leq i \leq n$. Then, by $\Sigma_{i=1}^{n}(\hat{Y}_i - \overline{Y})^2$ define the explained variance of the variable Y by the variable X .

   For the three variables, X, W and Y and for $n \geq 1$, let X_i, W_i, Y_i, $1 \leq i \leq n$. It is given that $\text{Corr}(Y, X) = 0.5$ and that $\text{Corr}(Y, W) = -0.6$. Suppose Y is regressed (separately) on $X, W, 2X, 2W, -3X$ and $-3W$. Then, each regression line comes with its corresponding explained variance of Y. Order these six values of the explained variance from the smallest to the largest. Note that ties are not ruled out.

   Solution:

   The explained variance of Y by X is $\text{Corr}^2(X, Y)\text{Var}(Y)$. Thus, for all six consider cases, the focus in on the square of the correlation. We know that $\text{Corr}(Y, 2X) = \text{Corr}(Y, X)$ and that $\text{Corr}(Y, -3X) = -\text{Corr}(Y, X)$. Thus using the date given, we know that

   $$\text{Corr}^2(Y, X) = \text{Corr}^2(Y, 2X) = \text{Corr}^2(Y, -3X) = 0.5^2 = 0.25.$$

   In a similar way

   $$\text{Corr}^2(Y, W) = \text{Corr}^2(Y, 2W) = \text{Corr}^2(Y, -3W) = 0.5^2 = 0.36^2.$$

   The order now follows.

3. Suppose for some parameter $\theta > 0$, a continuous random variable X that comes with a density function whose kernel equals $x^{\theta-1}$, $0 \leq x \leq 1$. Otherwise, the kernel equals zero.

   (a) (8) what is the density function of X?

   (b) (8) what is the k-th moment of X for $k \geq 1$?

   (c) (8) what is the variance of X?

   (d) (8) based on a random sample X_i, $1 \leq i \leq n$, all of which are independent and distributed as X, what is the MLE of θ?

   (e) (8) prove that the MLE for $e^{-\frac{1}{\theta}}$ is the geometric mean of the sample.

   (f) (8) what is the estimator of θ generated by the method of moments, when the k-th moment of the empirical distribution is used, $k \geq 1$?

   (g) (8) denote by $\hat{\theta}_k$ the k-th estimator from the previous item. Construct the estimator which is based on the weighted average among $\hat{\theta}_k$, $1 \leq k \leq 3$, where $\hat{\theta}_k$ receives a weight which is inversely proportional to k, $1 \leq k \leq 3$.

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Solution:

(a)

$$\int_{x=0}^{\infty} x^{\theta-1}\,dx = \frac{1}{\theta}x^{\theta}|_{x=0}^{1} = \frac{1}{\theta}$$

Hence,

$$\theta x^{\theta-1}, \quad 0 \leq x \leq 1,$$

is a density function.

(b)

$$\mathrm{E}(X^k) = \int_{x=0}^{1} x^k \theta x^{\theta-1}\,dx = \int_{x=0}^{1} \theta x^{k+\theta-1}\,dx$$

$$\theta \frac{x^{k+\theta}}{k+\theta}|_{x=0}^{1} = \frac{\theta}{k+\theta}.$$

(c)

$$\mathrm{Var}(X) = \mathrm{E}(X^2) - \mathrm{E}^2(X) = \frac{\theta}{2+\theta} - \frac{\theta^2}{(1+\theta)^2}$$

$$\frac{\theta}{(2+\theta)(1+\theta)^2}.$$

(d)

$$L(X_1, \ldots, X_n : \theta) = \Pi_{i=1}^{n} \theta X_i^{\theta-1} = \theta^n (\Pi_{i=1}^{n} X_i)^{\theta-1}.$$

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Then,

$$\log L(X_1, \ldots, X_n : \theta) = n \log \theta + (\theta - 1) \log \Pi_{i=1}^n X_i$$

and

$$\frac{d}{d\theta} \log L(X_1, \ldots, X_n : \theta) = \frac{n}{\theta} + \log \Pi_{i=1}^n X_i.$$

Equating this derivative to zero and solving for the MLE, we get that

$$\hat{\theta} = -\frac{n}{\log \Pi_{i=1}^n X_i} = -\frac{1}{\frac{1}{n} \log \Pi_{i=1}^n X_i} = -\frac{1}{\log (\Pi_{i=1}^n X_i)^{\frac{1}{n}}}.$$

(e) Since the MLE of the function is the function of the MLE, we get that the MLE for $e^{-1/\theta}$ equals $e^{-1/\hat{\theta}}$, which in fact equals

$$e^{\log (\Pi_{i=1}^n X_i)^{1/n}}$$

$$= (\Pi_{i=1}^n X_i)^{1/n},$$

which is the geometric mean of the series $X_1, \ldots, X_n$.

(f) Denote by $\hat{\theta}_k$ the estimator for θ which is based on the k-th moment for θ. Thus, by item (b) above

$$\overline{X^k} = \frac{\hat{\theta}_k}{k + \hat{\theta}_k}.$$

Solving for $\hat{\theta}_k$, we get that

$$\hat{\theta}_k = \frac{\overline{X^k}}{1 - \overline{X^k}}.$$

(g)

$$1 + \frac{1}{2} + \frac{1}{3} = \frac{11}{6}.$$

Thus the estimator we are after is

$$\frac{6}{11}(\hat{\theta}_1 + \frac{1}{2}\hat{\theta}_2 + \frac{1}{3}\hat{\theta}_3).$$

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